

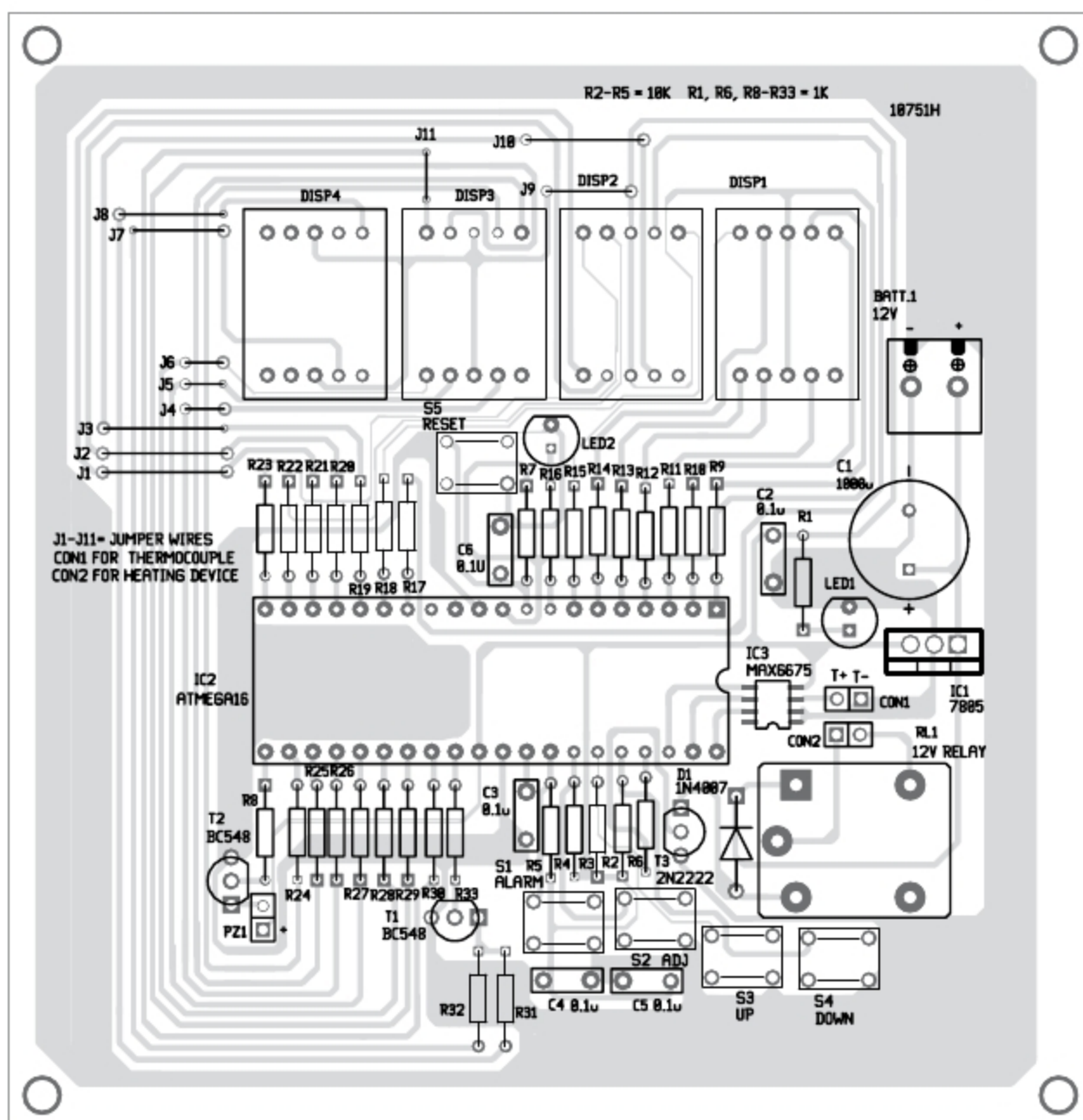


Fig. 3: Components layout for the PCB

start blinking and LED2 stops blinking, it indicates that the system is in edit mode.

Now, press S3 for value increments and S4 for decrements. Alarm value will change in steps of 10 units and temperature in steps of one unit. If there is no button pressed for more than 10 seconds, the current displayed value will be saved to EEPROM of the MCU, and the system will exit edit mode and display the current temperature value.

During testing of the circuit, note

the following:

1. Polarities of K-type thermocouple should be matched with MAX6675 polarity markings (T+ and T-).
2. MCU board and MAX6675 should be kept at a distance from the heating zone.
3. Accuracy of temperature and response time of change of temperature are mostly dependent on the thermocouple.
4. MAX6675 is more accurate up to 700°C (refer data sheet).
5. Press switch S5 momentarily

to reset, if there is a problem with the display.

With this project, you can adjust the accuracy of temperature readings to $\pm 25^\circ\text{C}$ with respect to a calibrated thermocouple. **EFY**



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EFY Note

The source code of this project is available for free download at source.efymag.com

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - ATmega16 8-bit AVR MCU
- IC3 - MAX6675 cold-junction-compensated K-thermocouple-to-digital converter
- D1 - 1N4007 rectifier diode
- LED1-LED2 - 5mm LED
- T1-T2 - BC548 npn transistor
- T3 - 2N2222 npn transistor
- DISP1-DISP4 - 7-segment display (common-anode)

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R6, R8-R33 - 1-kilo-ohm
- R2-R5, R7 - 10-kilo-ohm

Capacitors:

- C1 - 1000 μF , 35V electrolytic
- C2-C6 - 0.1 μF ceramic disk

Miscellaneous:

- PZ1 - Piezo buzzer
- S1-S5 - Tactile switch
- CON1 - 2-pin terminal connector for thermocouple
- CON2 - 2-pin connector for heating device
- BATT.1 - 12V battery
- RL1 - 12V single-changeover relay
- K-type thermocouple
- 40-pin IC socket
- Heatsink for IC1

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